

Probabilistic Forecasting

Multi-step Forecasting, Parametric Forecasts, Quantile Regression, and Evaluation

1 Multi-step Ahead Forecasting

Let $(x_t)_{t \geq 1}$ be a univariate or multivariate time series. At time t , we observe the past

$$x_{1:t} := (x_1, \dots, x_t). \quad (1)$$

The forecasting task is to use this observed past to predict future values.

1.1 Single-step versus multi-step forecasting

Single-step forecasting predicts only the next value:

$$\hat{x}_{t+1} = f_\theta(x_{1:t}). \quad (2)$$

The target is x_{t+1} .

Multi-step forecasting predicts an entire future horizon:

$$x_{t+1}, x_{t+2}, \dots, x_{t+H}, \quad (3)$$

where H is the forecasting horizon. The goal is to produce

$$\hat{x}_{t+1:t+H} := (\hat{x}_{t+1}, \dots, \hat{x}_{t+H}) \approx (x_{t+1}, \dots, x_{t+H}). \quad (4)$$

1.2 Direct forecasting

In **direct forecasting**, the model outputs all H future values at once:

$$f_\theta(x_{1:t}) = (\hat{x}_{t+1}, \dots, \hat{x}_{t+H}). \quad (5)$$

Equivalently,

$$[\hat{x}_{t+1}, \dots, \hat{x}_{t+H}] = f_\theta(x_1, \dots, x_t). \quad (6)$$

A common deterministic loss is

$$\mathcal{L}(\theta) = \sum_{h=1}^H \|x_{t+h} - \hat{x}_{t+h}\|_2^2. \quad (7)$$

For a dataset of forecasting windows indexed by $i = 1, \dots, N$:

$$\mathcal{L}(\theta) = \frac{1}{NH} \sum_{i=1}^N \sum_{h=1}^H \|x_{t_i+h}^{(i)} - \hat{x}_{t_i+h}^{(i)}\|_2^2. \quad (8)$$

Advantages. Direct forecasting has no mechanical error accumulation across horizons because $\hat{x}_{t+h}$ is not recursively fed into the prediction of $\hat{x}_{t+h+1}$. It is also fast at inference time: one forward pass gives all horizons.

Disadvantage. The model must learn all horizons simultaneously. Predicting x_{t+1} and predicting x_{t+H} may be very different tasks, especially when H is large.

1.3 Autoregressive forecasting

In **autoregressive forecasting**, a one-step model is reused repeatedly. First:

$$\hat{x}_{t+1} = f_{\theta}(x_1, \dots, x_t). \quad (9)$$

Then the model feeds its own prediction back as input:

$$\hat{x}_{t+2} = f_{\theta}(x_2, \dots, x_t, \hat{x}_{t+1}), \quad (10)$$

$$\hat{x}_{t+3} = f_{\theta}(x_3, \dots, x_t, \hat{x}_{t+1}, \hat{x}_{t+2}), \quad (11)$$

and so on until $\hat{x}_{t+H}$.

If the model uses a context window of length L , the recursion can be written as

$$\hat{x}_{t+h} = f_{\theta}(\tilde{x}_{t+h-L}, \dots, \tilde{x}_{t+h-1}), \quad (12)$$

where

$$\tilde{x}_s = \begin{cases} x_s, & s \leq t, \\ \hat{x}_s, & s > t. \end{cases} \quad (13)$$

Advantages. The same model can be reused for variable horizons. If one wants $H = 10$ instead of $H = 5$, the rollout is simply continued for more steps.

Disadvantages. Errors accumulate. If $\hat{x}_{t+1}$ is wrong, then $\hat{x}_{t+2}$ is computed using a corrupted input. Moreover, there is exposure bias: during training the model often sees clean ground-truth histories, but during inference it sees its own imperfect predictions.

2 Training Strategies for Autoregressive Multi-step Forecasting

2.1 Strategy 1: next-step training and autoregressive inference

The simplest strategy is to train the model only for one-step prediction:

$$\hat{x}_{t+1} = f_{\theta}(x_1, \dots, x_t). \quad (14)$$

The training objective is

$$\mathcal{L}(\theta) = \frac{1}{T} \sum_{t=1}^T \|x_{t+1} - \hat{x}_{t+1}\|_2^2. \quad (15)$$

At inference time, the model is rolled out autoregressively:

$$\hat{x}_{t+1} = f_{\theta}(x_1, \dots, x_t), \quad (16)$$

$$\hat{x}_{t+2} = f_{\theta}(x_2, \dots, x_t, \hat{x}_{t+1}), \quad (17)$$

$$\hat{x}_{t+3} = f_{\theta}(x_3, \dots, x_t, \hat{x}_{t+1}, \hat{x}_{t+2}). \quad (18)$$

Key limitation

This strategy is simple, but it creates a train-test mismatch. During training, the model sees true values as inputs. During inference, it sees its own predictions. This mismatch is the source of exposure bias and error accumulation.

2.2 Strategy 2: curriculum learning

Curriculum learning makes the training task progressively harder. Instead of training immediately on the full horizon H , the model starts with shorter horizons and gradually increases the rollout length.

For example:

epochs 1-10 : predict 1 step ahead, (19)

epochs 11-20 : predict 2 steps ahead autoregressively, (20)

epochs 21-30 : predict 3 steps ahead autoregressively, (21)

and so on until the full horizon H is reached.

Let H_e denote the horizon used at epoch e , with

$$1 \leq H_e \leq H. \quad (22)$$

The epoch-dependent training loss is

$$\mathcal{L}_e(\theta) = \frac{1}{NH_e} \sum_{i=1}^N \sum_{h=1}^{H_e} \|x_{t_i+h}^{(i)} - \hat{x}_{t_i+h}^{(i)}\|_2^2. \quad (23)$$

The purpose is to expose the model progressively to its own predictions. The model first learns short rollouts, then longer ones. This helps bridge the gap between teacher-forced training and free-running inference.

2.3 Strategy 3: scheduled sampling

Scheduled sampling mixes ground-truth values and model predictions during training.

At a rollout step, suppose the model has predicted $\hat{x}_{t+h}$. The next input is chosen as

$$x_{t+h}^{\text{input}} = \begin{cases} x_{t+h}, & \text{with probability } p, \\ \hat{x}_{t+h}, & \text{with probability } 1 - p. \end{cases} \quad (24)$$

At the beginning of training, $p \approx 1$, so the model mostly receives ground-truth values. Later, p decreases toward 0, so the model increasingly receives its own predictions.

A simple linear schedule is

$$p_e = \max\left(0, 1 - \frac{e}{E}\right), \quad (25)$$

where e is the epoch and E is the number of epochs over which teacher forcing is annealed. For example, if $E = 100$, then

$$p_e = 1 - \frac{e}{100} \quad (26)$$

for $0 \leq e \leq 100$.

The target remains the true future trajectory. Only the rollout inputs are randomly chosen. The multi-step loss is still

$$\mathcal{L}(\theta) = \sum_{h=1}^H \|x_{t+h} - \hat{x}_{t+h}\|_2^2. \quad (27)$$

Interpretation

Scheduled sampling gradually turns training from teacher-forced mode into inference-like autoregressive mode. Early training is stable because the model receives true inputs; later training becomes more realistic because the model learns to handle its own errors.

3 From Point Forecasting to Probabilistic Forecasting

A deterministic forecasting model outputs a single value:

$$\hat{x}_{t+h} = f_{\theta}(x_{1:t}). \quad (28)$$

But the future is uncertain. Two forecasts can have the same central value but very different uncertainty:

$$X_{t+h} \sim \mathcal{N}(100, 2^2) \quad \text{versus} \quad X_{t+h} \sim \mathcal{N}(100, 20^2). \quad (29)$$

Both have mean 100, but the second is much less certain.

Probabilistic forecasting predicts a conditional distribution:

$$p(x_{t+h} \mid x_1, \dots, x_t). \quad (30)$$

This allows us to quantify uncertainty, build prediction intervals, make risk-aware decisions, and handle non-Gaussian or asymmetric uncertainty.

4 Parametric Probabilistic Forecasting

4.1 Model formulation

The parametric approach chooses a distribution family and makes the neural network output the parameters of that family. For example, choose a Gaussian distribution:

$$X_{t+h} \mid x_{1:t} \sim \mathcal{N}(\mu_{t+h}, \sigma_{t+h}^2). \quad (31)$$

The neural network outputs

$$(\hat{\mu}_{t+h}, s_{t+h}) = f_{\theta}(x_{1:t}), \quad (32)$$

where $s_{t+h} \in \mathbb{R}$ is unconstrained. Since the standard deviation must be positive, define

$$\hat{\sigma}_{t+h} = \exp(s_{t+h}) \quad (33)$$

or

$$\hat{\sigma}_{t+h} = \text{softplus}(s_{t+h}) + \varepsilon, \quad (34)$$

with a small $\varepsilon > 0$ for numerical stability.

Then the predictive distribution is

$$X_{t+h} \mid x_{1:t} \sim \mathcal{N}(\hat{\mu}_{t+h}, \hat{\sigma}_{t+h}^2). \quad (35)$$

4.2 What are $\hat{\mu}$ and $\hat{\sigma}$?

In this neural forecasting setting, $\hat{\mu}$ and $\hat{\sigma}$ are not global closed-form estimates. They are outputs of a neural network, hence functions of the observed history:

$$\hat{\mu}_{t+h} = \mu_{\theta}(x_{1:t}), \quad \hat{\sigma}_{t+h} = \sigma_{\theta}(x_{1:t}). \quad (36)$$

The network parameters θ are learned by gradient descent.

This differs from classical i.i.d. Gaussian maximum likelihood. If

$$y_1, \dots, y_N \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2), \quad (37)$$

then one has closed-form estimates

$$\hat{\mu} = \frac{1}{N} \sum_{i=1}^N y_i, \quad \hat{\sigma}^2 = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{\mu})^2. \quad (38)$$

In probabilistic forecasting, however, the parameters are conditional on the input history, and the conditional mapping is learned by optimizing θ .

4.3 Gaussian negative log-likelihood derivation

For simplicity, write

$$y = x_{t+h}, \quad \hat{\mu} = \hat{\mu}_{t+h}, \quad \hat{\sigma} = \hat{\sigma}_{t+h}. \quad (39)$$

The Gaussian density is

$$p_{\theta}(y \mid x_{1:t}) = \frac{1}{\sqrt{2\pi\hat{\sigma}^2}} \exp\left(-\frac{(y - \hat{\mu})^2}{2\hat{\sigma}^2}\right). \quad (40)$$

The log-likelihood is

$$\log p_{\theta}(y \mid x_{1:t}) = \log \left[\frac{1}{\sqrt{2\pi\hat{\sigma}^2}} \exp\left(-\frac{(y - \hat{\mu})^2}{2\hat{\sigma}^2}\right) \right] \quad (41)$$

$$= \log\left(\frac{1}{\sqrt{2\pi\hat{\sigma}^2}}\right) - \frac{(y - \hat{\mu})^2}{2\hat{\sigma}^2} \quad (42)$$

$$= -\frac{1}{2} \log(2\pi\hat{\sigma}^2) - \frac{(y - \hat{\mu})^2}{2\hat{\sigma}^2}. \quad (43)$$

Therefore, the negative log-likelihood is

$$\boxed{\ell(\theta) = -\log p_{\theta}(y \mid x_{1:t}) = \frac{1}{2} \log(2\pi\hat{\sigma}^2) + \frac{(y - \hat{\mu})^2}{2\hat{\sigma}^2}.} \quad (44)$$

Returning to time-series notation:

$$\boxed{\ell_{t,h}(\theta) = \frac{1}{2} \log(2\pi\hat{\sigma}_{t+h}^2) + \frac{(x_{t+h} - \hat{\mu}_{t+h})^2}{2\hat{\sigma}_{t+h}^2}.} \quad (45)$$

4.4 Interpretation of the two terms

The loss has two components:

$$\ell = \underbrace{\frac{1}{2} \log(2\pi\hat{\sigma}^2)}_{\text{uncertainty penalty}} + \underbrace{\frac{(y - \hat{\mu})^2}{2\hat{\sigma}^2}}_{\text{normalized prediction error}}. \quad (46)$$

The second term is a squared error divided by the predicted variance. A large error is less penalized if the model predicted large uncertainty. But the first term penalizes large variance through $\log(\hat{\sigma}^2)$. Thus the model cannot make all variances infinite.

If $\hat{\sigma}^2 \rightarrow 0$ and $y \neq \hat{\mu}$, then

$$\frac{(y - \hat{\mu})^2}{2\hat{\sigma}^2} \rightarrow +\infty. \quad (47)$$

If $\hat{\sigma}^2 \rightarrow +\infty$, then

$$\frac{1}{2} \log(2\pi\hat{\sigma}^2) \rightarrow +\infty. \quad (48)$$

So the NLL encourages uncertainty to be large only when needed.

4.5 Dataset loss: one horizon and multi-step horizon

For one horizon h , given N training windows,

$$\mathcal{D} = \{(x_{1:t}^{(i)}, x_{t+h}^{(i)})\}_{i=1}^N, \quad (49)$$

the model outputs

$$(\hat{\mu}_{i,h}, \hat{\sigma}_{i,h}) = f_{\theta}(x_{1:t}^{(i)}). \quad (50)$$

The dataset loss is

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^N \left[\frac{1}{2} \log(2\pi\hat{\sigma}_{i,h}^2) + \frac{(x_{t+h}^{(i)} - \hat{\mu}_{i,h})^2}{2\hat{\sigma}_{i,h}^2} \right]. \quad (51)$$

For multi-step forecasting, suppose the model outputs

$$\{(\hat{\mu}_{i,h}, \hat{\sigma}_{i,h})\}_{h=1}^H. \quad (52)$$

If we assume a factorized likelihood over horizons,

$$p_{\theta}(x_{t+1:t+H}^{(i)} | x_{1:t}^{(i)}) = \prod_{h=1}^H \mathcal{N}(x_{t+h}^{(i)}; \hat{\mu}_{i,h}, \hat{\sigma}_{i,h}^2), \quad (53)$$

then

$$\mathcal{L}(\theta) = \frac{1}{NH} \sum_{i=1}^N \sum_{h=1}^H \left[\frac{1}{2} \log(2\pi\hat{\sigma}_{i,h}^2) + \frac{(x_{t+h}^{(i)} - \hat{\mu}_{i,h})^2}{2\hat{\sigma}_{i,h}^2} \right]. \quad (54)$$

4.6 Prediction intervals

Since the model predicts a distribution, it naturally gives prediction intervals. For a Gaussian forecast,

$$X_{t+h} \mid x_{1:t} \sim \mathcal{N}(\hat{\mu}_{t+h}, \hat{\sigma}_{t+h}^2), \quad (55)$$

a $100(1 - \alpha)\%$ prediction interval is

$$\left[\hat{\mu}_{t+h} - z_{1-\alpha/2} \hat{\sigma}_{t+h}, \hat{\mu}_{t+h} + z_{1-\alpha/2} \hat{\sigma}_{t+h} \right], \quad (56)$$

where $z_{1-\alpha/2}$ is a standard normal quantile. For example,

$$90\% : \hat{\mu} \pm 1.645 \hat{\sigma}, \quad (57)$$

$$95\% : \hat{\mu} \pm 1.96 \hat{\sigma}, \quad (58)$$

$$99\% : \hat{\mu} \pm 2.576 \hat{\sigma}. \quad (59)$$

These are prediction intervals for future values, not confidence intervals for a fixed parameter.

5 Quantile Regression

5.1 Motivation

The Gaussian parametric model is convenient, but it assumes a symmetric unimodal distribution. Quantile regression avoids choosing a full parametric distribution family. Instead, the model directly predicts quantiles of the future value.

For example, the network may output

$$f_\theta(x_{1:t}) = (\hat{q}_{0.1,t+h}, \hat{q}_{0.5,t+h}, \hat{q}_{0.9,t+h}). \quad (60)$$

Here $\hat{q}_{0.1}$, $\hat{q}_{0.5}$, and $\hat{q}_{0.9}$ estimate the 10th, 50th, and 90th percentiles of

$$X_{t+h} \mid x_{1:t}. \quad (61)$$

5.2 Definition of a quantile

Let Y be a random variable with cumulative distribution function

$$F(y) = \mathbb{P}(Y \leq y). \quad (62)$$

The τ -quantile is

$$q_\tau = \inf\{q \in \mathbb{R} : F(q) \geq \tau\}. \quad (63)$$

Equivalently, one should have approximately

$$\mathbb{P}(Y \leq q_\tau) \approx \tau. \quad (64)$$

For forecasting,

$$q_{\tau,t+h} \quad (65)$$

is the conditional τ -quantile of X_{t+h} given $x_{1:t}$.

5.3 Prediction intervals from quantiles

If the model predicts $\hat{q}_{0.1,t+h}$ and $\hat{q}_{0.9,t+h}$, then

$$[\hat{q}_{0.1,t+h}, \hat{q}_{0.9,t+h}] \quad (66)$$

is an approximate 80% prediction interval, since $0.9 - 0.1 = 0.8$. More generally,

$$[\hat{q}_{\alpha/2,t+h}, \hat{q}_{1-\alpha/2,t+h}] \quad (67)$$

is an approximate $100(1 - \alpha)\%$ interval.

Unlike Gaussian intervals, quantile intervals can be asymmetric:

$$\hat{q}_{0.5} - \hat{q}_{0.1} \neq \hat{q}_{0.9} - \hat{q}_{0.5}. \quad (68)$$

This is useful when the future distribution is skewed.

5.4 Pinball loss

Let $y = x_{t+h}$ be the true future value and let $\hat{q}_\tau$ be the predicted τ -quantile. Define

$$u = y - \hat{q}_\tau. \quad (69)$$

The quantile loss, also called pinball loss, is

$$\rho_\tau(u) = \begin{cases} \tau u, & u \geq 0, \\ (\tau - 1)u, & u < 0. \end{cases} \quad (70)$$

Equivalently,

$$\boxed{\rho_\tau(y - \hat{q}_\tau) = \max\{\tau(y - \hat{q}_\tau), (\tau - 1)(y - \hat{q}_\tau)\}.} \quad (71)$$

If $\hat{q}_\tau < y$, then the model underpredicts and

$$\rho_\tau(y - \hat{q}_\tau) = \tau(y - \hat{q}_\tau). \quad (72)$$

If $\hat{q}_\tau > y$, then the model overpredicts and

$$\rho_\tau(y - \hat{q}_\tau) = (1 - \tau)(\hat{q}_\tau - y). \quad (73)$$

Thus the penalty is asymmetric.

For $\tau = 0.9$, underprediction is penalized with weight 0.9 while overprediction is penalized with weight 0.1. Hence the model is pushed upward until roughly 90% of observations fall below the predicted value. For $\tau = 0.1$, the opposite happens. For $\tau = 0.5$, the pinball loss becomes proportional to absolute error:

$$\rho_{0.5}(y - \hat{q}) = \frac{1}{2}|y - \hat{q}|, \quad (74)$$

so the 0.5-quantile is the median.

5.5 Dataset loss

For one quantile and one horizon, the dataset loss is

$$\mathcal{L}_\tau(\theta) = \frac{1}{N} \sum_{i=1}^N \rho_\tau(y_i - \hat{q}_{\tau,i}). \quad (75)$$

For multiple quantiles $\mathcal{Q} = \{\tau_1, \dots, \tau_K\}$,

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^N \sum_{\tau \in \mathcal{Q}} \rho_\tau(y_i - \hat{q}_{\tau,i}). \quad (76)$$

For multi-step forecasting,

$$\mathcal{L}(\theta) = \frac{1}{NH} \sum_{i=1}^N \sum_{h=1}^H \sum_{\tau \in \mathcal{Q}} \rho_\tau(x_{t_i+h}^{(i)} - \hat{q}_{\tau,t_i+h}^{(i)}). \quad (77)$$

5.6 Quantile crossing

A practical issue is quantile crossing. Ideally,

$$\hat{q}_{0.1} \leq \hat{q}_{0.5} \leq \hat{q}_{0.9}. \quad (78)$$

If the network outputs quantiles independently, this ordering can be violated. A common fix is a monotone parameterization:

$$\hat{q}_{0.1} = a, \quad (79)$$

$$\hat{q}_{0.5} = a + \text{softplus}(b), \quad (80)$$

$$\hat{q}_{0.9} = a + \text{softplus}(b) + \text{softplus}(c). \quad (81)$$

This guarantees non-crossing quantiles.

6 Evaluation of Probabilistic Forecasts

Probabilistic forecasts should be evaluated not only by their central accuracy, but also by the quality of their uncertainty. The main concepts are calibration, sharpness, and proper scoring rules.

6.1 Calibration

Calibration asks whether predicted probabilities match empirical frequencies. Suppose the model gives a 90% prediction interval

$$I_i^{90} = [L_i, U_i]. \quad (82)$$

A calibrated model should satisfy

$$\mathbb{P}(Y_i \in I_i^{90}) \approx 0.9. \quad (83)$$

On a test set, the empirical coverage is

$$\widehat{\text{Coverage}}_{90} = \frac{1}{N} \sum_{i=1}^N \mathbb{1}\{y_i \in [L_i, U_i]\}. \quad (84)$$

The model is calibrated at level 90% if

$$\widehat{\text{Coverage}}_{90} \approx 0.9. \quad (85)$$

For Gaussian parametric forecasts, a 90% interval is

$$[\hat{\mu}_i - 1.645\hat{\sigma}_i, \hat{\mu}_i + 1.645\hat{\sigma}_i]. \quad (86)$$

For quantile regression, a 90% interval is

$$[\hat{q}_{0.05,i}, \hat{q}_{0.95,i}]. \quad (87)$$

For a single quantile, calibration means

$$\frac{1}{N} \sum_{i=1}^N \mathbb{1}\{y_i \leq \hat{q}_{\tau,i}\} \approx \tau. \quad (88)$$

6.2 Sharpness

Sharpness measures how concentrated or narrow the forecasts are. Among well-calibrated forecasts, narrower intervals are preferred.

For a 90% interval $I_i^{90} = [L_i, U_i]$, average width is

$$\text{Width}_{90} = \frac{1}{N} \sum_{i=1}^N (U_i - L_i). \quad (89)$$

For Gaussian forecasts,

$$U_i - L_i = 2(1.645)\hat{\sigma}_i. \quad (90)$$

For quantile forecasts,

$$U_i - L_i = \hat{q}_{0.95,i} - \hat{q}_{0.05,i}. \quad (91)$$

Sharpness alone is not sufficient: intervals can be narrow but badly calibrated. The desired property is sharpness subject to calibration.

6.3 Log-likelihood on held-out data

If the model provides a full predictive density, one can evaluate the held-out log-likelihood:

$$\text{LL} = \sum_{i=1}^N \log p_{\theta}(y_i \mid x_{1:t_i}^{(i)}). \quad (92)$$

Equivalently, minimize

$$\text{NLL} = - \sum_{i=1}^N \log p_{\theta}(y_i \mid x_{1:t_i}^{(i)}). \quad (93)$$

For Gaussian forecasts,

$$\text{NLL} = \sum_{i=1}^N \left[\frac{1}{2} \log(2\pi\hat{\sigma}_i^2) + \frac{(y_i - \hat{\mu}_i)^2}{2\hat{\sigma}_i^2} \right]. \quad (94)$$

Log-likelihood is natural for parametric probabilistic models, but it is not directly available for basic quantile regression because quantile regression does not output a full density.

6.4 Proper scoring rules

A scoring rule evaluates a probabilistic forecast after the true value is observed. It is proper if the expected score is optimized by reporting the true predictive distribution. In simple terms, a proper scoring rule encourages honest uncertainty.

For parametric probabilistic models, common proper scoring rules include negative log-likelihood and CRPS. For quantile regression, the pinball loss is a proper scoring rule for quantiles.

7 Continuous Ranked Probability Score

7.1 Definition

Let F_i be the predicted CDF for the target Y_i :

$$F_i(z) = \mathbb{P}_\theta(Y_i \leq z \mid x_{1:t_i}^{(i)}). \quad (95)$$

Let the observed true value be y_i . The ideal distribution after observing the truth is a Dirac mass at y_i , denoted δ_{y_i} . Its CDF is the step function

$$F_{\delta_{y_i}}(z) = \mathbb{1}\{z \geq y_i\}. \quad (96)$$

The CRPS is

$$\boxed{\text{CRPS}(F_i, y_i) = \int_{-\infty}^{+\infty} (F_i(z) - \mathbb{1}\{z \geq y_i\})^2 \, dz.} \quad (97)$$

Thus CRPS is the area between the predicted CDF and the ideal CDF concentrated at the true value. Lower CRPS is better.

7.2 Dirac mass recap

A Dirac distribution at y , denoted δ_y , is a distribution with all probability mass at y :

$$\mathbb{P}(X = y) = 1. \quad (98)$$

Its CDF is

$$F_{\delta_y}(z) = \begin{cases} 0, & z < y, \\ 1, & z \geq y. \end{cases} \quad (99)$$

In CRPS, the Dirac distribution is the perfect forecast after the outcome has been observed.

7.3 Why CRPS generalizes MAE

If the forecast is deterministic, it can be represented as a Dirac mass at $\hat{y}$. Its CDF is

$$F(z) = \mathbb{1}\{z \geq \hat{y}\}. \quad (100)$$

Then the CRPS becomes

$$\text{CRPS}(F, y) = |\hat{y} - y|. \quad (101)$$

So CRPS reduces to MAE for deterministic point forecasts and generalizes MAE to full distributions.

7.4 Closed-form CRPS for a Gaussian forecast

If

$$F = \mathcal{N}(\mu, \sigma^2), \quad (102)$$

then

$$\boxed{\text{CRPS}(F, y) = \sigma \left[z(2\Phi(z) - 1) + 2\phi(z) - \frac{1}{\sqrt{\pi}} \right]}, \quad (103)$$

where

$$z = \frac{y - \mu}{\sigma}, \quad (104)$$

Φ is the standard normal CDF, and ϕ is the standard normal density.

7.5 Gaussian CRPS example

Suppose the model predicts

$$Y \mid x_{1:t} \sim \mathcal{N}(100, 10^2), \quad (105)$$

and the true observed value is

$$y = 112. \quad (106)$$

Then

$$z = \frac{112 - 100}{10} = 1.2. \quad (107)$$

Using

$$\Phi(1.2) \approx 0.8849, \quad \phi(1.2) \approx 0.1942, \quad \frac{1}{\sqrt{\pi}} \approx 0.5642, \quad (108)$$

we get

$$\text{CRPS} = 10 [1.2(2 \cdot 0.8849 - 1) + 2(0.1942) - 0.5642] \quad (109)$$

$$= 10 [1.2(0.7698) + 0.3884 - 0.5642] \quad (110)$$

$$= 10(0.7480) \quad (111)$$

$$\approx 7.48. \quad (112)$$

The point absolute error is

$$|112 - 100| = 12. \quad (113)$$

CRPS is smaller because the model predicted a distribution with nonzero uncertainty, and the value 112 was plausible under $\mathcal{N}(100, 10^2)$.

7.6 Effect of bad uncertainty

Keeping the same mean $\mu = 100$ and observed value $y = 112$:

Forecast	Interpretation	Approximate CRPS
$\mathcal{N}(100, 2^2)$	too confident	10.87
$\mathcal{N}(100, 10^2)$	reasonable uncertainty	7.48
$\mathcal{N}(100, 30^2)$	too uncertain	8.90

Thus CRPS penalizes both bad location and bad uncertainty spread.

8 Which Evaluation Metric for Which Model?

Metric	Parametric model	probabilistic	Quantile regression
Calibration	Yes: interval coverage from the distribution		Yes: quantile coverage or interval coverage
Sharpness	Yes: interval width, often controlled by $\hat{\sigma}$		Yes: quantile interval width
Log-likelihood / NLL	Natural, because the density is available		Not direct, unless a density is reconstructed
CRPS	Natural if the CDF is available		Possible approximately if many quantiles reconstruct a CDF
Pinball loss	Not the main metric		Natural training and evaluation score

Final takeaway

Parametric probabilistic models output a full distribution, so likelihood-based metrics and CRPS are natural. Quantile regression outputs selected quantiles, so pinball loss, calibration, and interval sharpness are the most direct evaluation tools. Both families should be checked for calibration and sharpness.